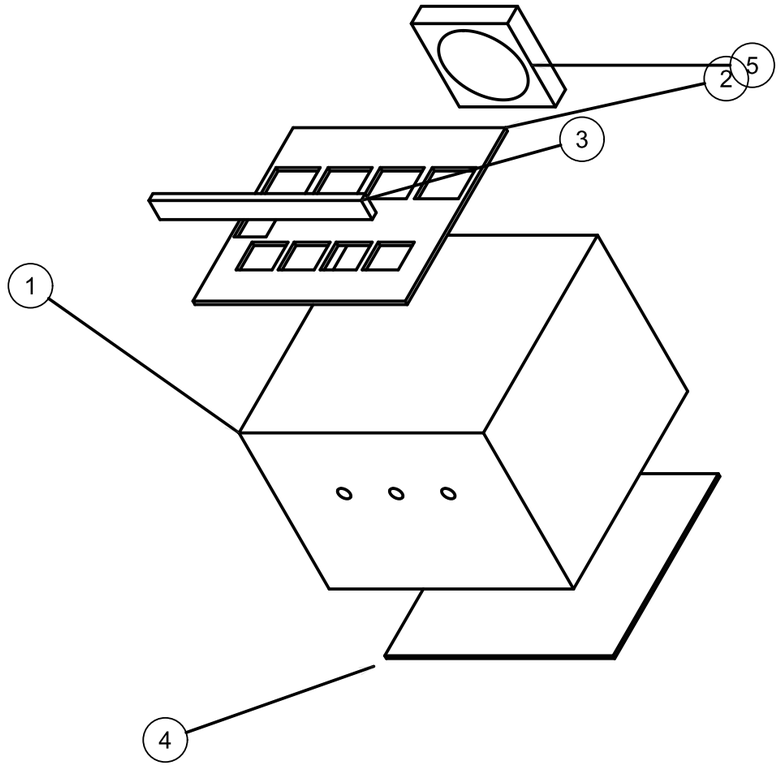
# ASSEMBLY – OpenSmell v1.1 sensing platform

exploded isometric (parts separated along the assembly axis) + general arrangement – shell, baffle plate, filter pack, PCB chamber floor, DIA40 exhaust fan

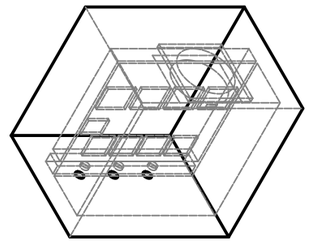
## BILL OF MATERIALS

| ITEM | PART / DESCRIPTION | QTY | MATERIAL / SOURCE | TYPE |
|---|---|---|---|---|
| 1 | Shell  122×114×90, cavity 106×100×48 | 1 | PETG FDM (or AL 6061-T6) | make |
| 2 | Baffle plate  106×100×2, 8 wells + ref port | 1 | PTFE machined (or PVDF) | make |
| 3 | Filter pack, full-lane series element | 1 | pleated PTFE, ≥70 cm² media | buy |
| 4 | Main PCB  114×105, sealed chamber floor | 1 | FR4 1.6 mm + build (OS-003) | make |
| 5 | Axial fan  DIA40 × t10, 5 V | 1 | off-the-shelf (any 40 mm) | buy |

geometry from real HLR of the v1.1 solid; all dims mm; per blueprint.md 0.4




EXPLODED – assembly order  (NTS)



ISO – general arrangement  (NTS)

NOTES
1. ENVELOPE 122 × 114 × 90; CAVITY 106 × 100 × 48.
2. FLOW: DIA6 INTAKE → FILTER → 11 mm LANE → 8 WELLS → DUCT → DIA40 FAN.
3. PCB = CHAMBER FLOOR (OS-001); CARTRIDGES LIFT OUT WITH THE PLATE.
4. NO SILICONE, PVC, PAPER OR FOAM WHERE THE SAMPLE TOUCHES.
5. ITEM NUMBERS MATCH THE BOM; EXPLODED IS A TRUE HLR.

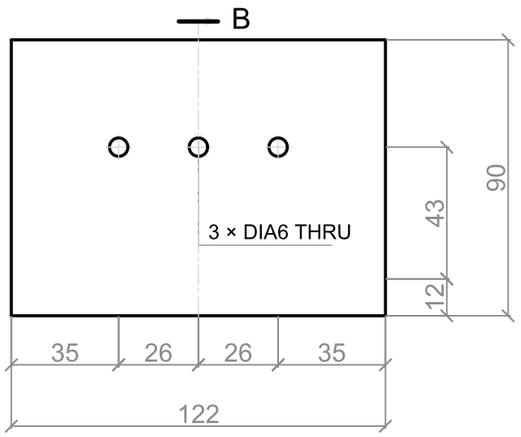
GENERAL TOLERANCE  ISO 2768-mK  ·  PROJECT OpenSmell v1.1  REV L
NOTES:  all dims mm; deburr/de-flash; seal wetted faces; print thin parts 0.2 mm layer

| | | | |
|---|---|---|---|
| ASSEMBLY | | | SCALE  NTS |
| DRAWING No.  OS-000 | REV L | SHEET 1/3 | SIZE  A3<br>UNITS  mm |
| MATERIAL  see BOM<br>FINISH  see BOM | | | FIRST ANGLE<br>(ISO 128) |
| DRAWN  OpenSmell | CHECK  – | DATE  2026-10-18 | |

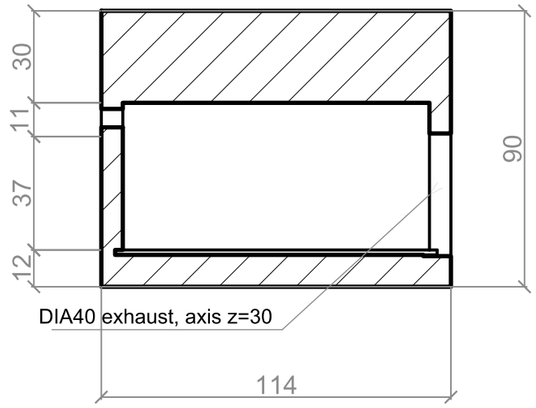
# SHELL – main body  (OS-001)

PETG FDM, printed as one part; the PCB is the sealed chamber floor and the baffle plate seats on the ledge at z=49

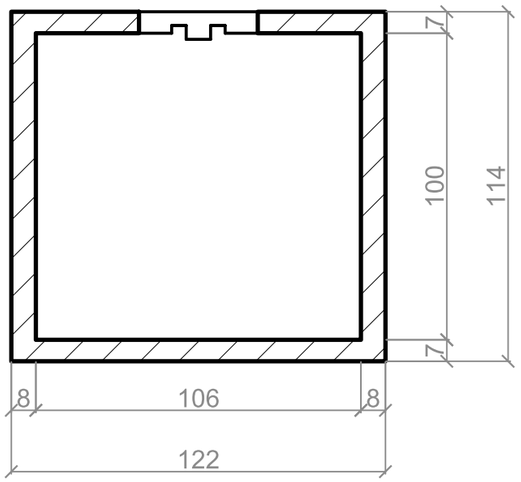
FRONT – intake face  1:2

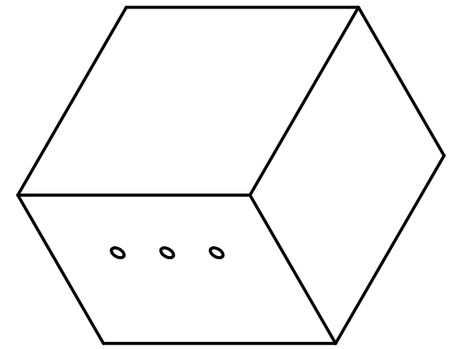



SECTION B-B – x=61  1:2




SECTION C-C – z=35  1:2



ISO – shell as printed  (NTS)

NOTES

1. MATERIAL: PETG FDM, 0.2 mm LAYER, 4 PERIMETERS; SEAL WETTED FACES.
2. THE PCB IS THE CHAMBER FLOOR: SEAT POCKET 1.6 DEEP, BOARD TOP z=12.
3. BAFFLE PLANE z=49 (37 ABOVE THE FLOOR). MODULE ALLOWANCE 29, RECESS 8.
4. FLOW: 3 × DIA6 INTAKE (z=55) → FILTER (y=14) → 11 mm LANE → 8 WELLS → DUCT → DIA40 FAN.
5. ALL OPENINGS DEBURRED; NO SILICONE, PVC, PAPER OR FOAM IN THE GAS PATH.
6. DIMS IN mm. DATUM: FRONT-LOWER-LEFT OUTER CORNER (0, 0, 0).

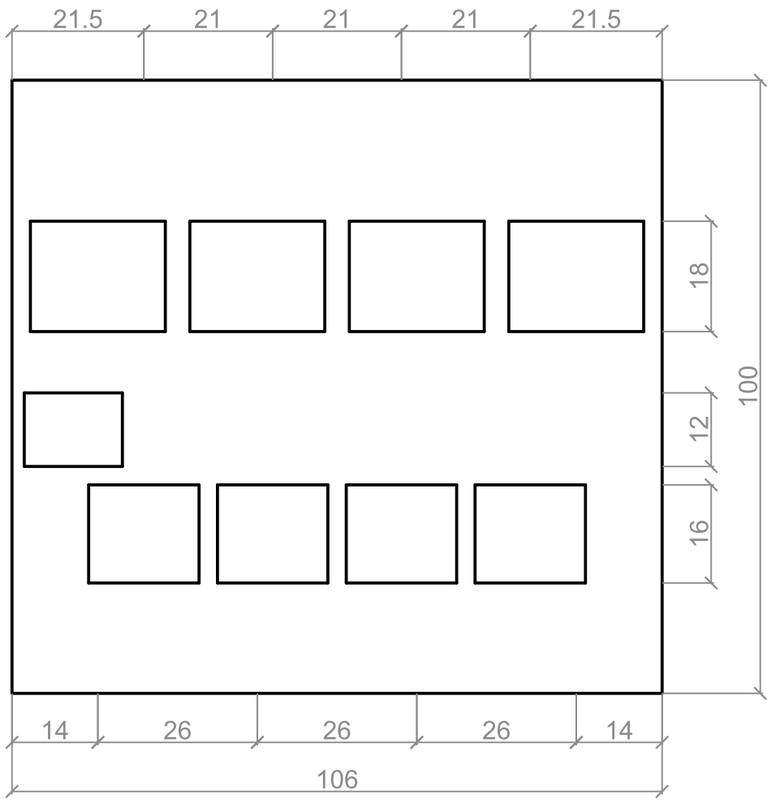
GENERAL TOLERANCE  ISO 2768-mK  ·  PROJECT OpenSmell v1.1  REV L
NOTES:  all dims mm; deburr/de-flash; seal wetted faces; print thin parts 0.2 mm layer

| | | | |
|---|---|---|---|
| SHELL – main body | | | SCALE  1:2 |
| DRAWING No.  OS-001 | REV L | SHEET 2/3 | SIZE  A3<br>UNITS  mm |
| MATERIAL  PETG, FDM 0.2 mm, 4 perimeters, vapor-sealed inner cavity<br>FINISH  as-printed, de-flash; wetted faces seamed/sealed | | | FIRST ANGLE<br>(ISO 128) |
| DRAWN  OpenSmell | CHECK  – | DATE  2026-10-18 | |

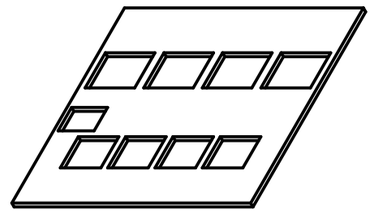
# BAFFLE – well plate  (OS-002)

PTFE plate that closes the chamber; every well is open-bottom, so each cell reads the volume under it

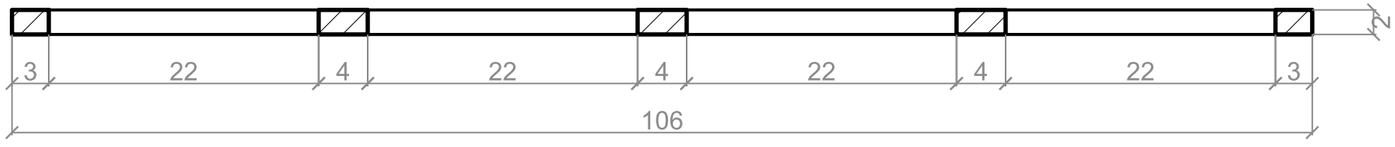
NOTES
1. MATERIAL: PTFE, MACHINED. SUBSTITUTE: PVDF OR BOROSILICATE.
2. PLATE 106 × 100 × 2, SEATS ON THE SHELL LEDGE AT z=47.
3. WELL ORDER IS FLOW ORDER: DIGITAL UPSTREAM, ANALOG DOWNSTREAM.
4. EVERY WELL IS OPEN-BOTTOM: THE CELL READS THE SUB-BAFFLE VOLUME.
5. ANALOG WELLS 22 × 18 (A1–A4), DIGITAL WELLS 18 × 16 (D1–D4).
6. U5 REFERENCE PORT 16 × 12 AT (18, 44): SHT31 SEES THE SAME AIR.
7. GENERAL TOLERANCE ISO 2768-mK; WETTED FACES FREE OF RESIDUE.




PLAN (8 wells + reference port)  1:1 – PRINT AT 100%



ISO – open-bottom wells  (NTS)




SECTION A-A – analog well row  2:1

GENERAL TOLERANCE  ISO 2768-mK  ·  PROJECT OpenSmell v1.1  REV L
NOTES:  all dims mm; deburr/de-flash; seal wetted faces; print thin parts 0.2 mm layer

| | | | |
|---|---|---|---|
| BAFFLE – well plate | | | SCALE  AS NOTED |
| DRAWING No.  OS-002 | REV L | SHEET 3/3 | SIZE  A3<br>UNITS  mm |
| MATERIAL  PTFE, machined (inert, non-porous, hydrophobic)<br>FINISH  deburr; 9 openings, edges broken 0.2 | | | FIRST ANGLE<br>(ISO 128) |
| DRAWN  OpenSmell | CHECK  – | DATE  2026-10-18 | |